A Comprehensive Global Guide to Disaster Preparedness, Response, and Recovery

Foundational Concepts in Disaster Management

1.0 Understanding Disaster Risk: A Global Framework

The modern understanding of disasters has evolved significantly, moving from a reactive focus on post-event response to a proactive strategy of risk reduction. This paradigm is built upon a precise, internationally recognized vocabulary and a comprehensive framework for action that addresses the full lifecycle of a disaster.

1.0.1 Defining "Disaster"

A foundational element of effective disaster management is a shared, precise definition of the term itself. The United Nations Office for Disaster Risk Reduction (UNDRR) provides the globally accepted definition, characterizing a disaster as: "A serious disruption of the functioning of a community or a society at any scale due to hazardous events interacting with conditions of exposure, vulnerability and capacity, leading to one or more of the following: human, material, economic and environmental losses and impacts".[1]

This definition is critical because it shifts the focus away from the hazard event alone (e.g., an earthquake or a flood) and toward the societal conditions that allow such an event to cause widespread harm. A disaster is not merely the occurrence of a natural phenomenon; it is the consequence of that phenomenon impacting a vulnerable population that lacks the capacity to cope. The effects can be immediate and localized but are often widespread and can persist for extended periods, frequently exceeding a community's ability to manage using its own resources.[1] This necessitates a structured approach to management that considers the interplay of these factors.

1.0.2 The Disaster Management Cycle

The management of disasters is conceptualized as a continuous cycle with distinct but interrelated phases. This model, referenced by major international organizations such as the American Red Cross, provides a structure for all actions taken before, during, and after an event.[2] The principal phases are:

- **Preparedness:** Actions taken prior to a disaster to ensure an effective response. This includes developing plans, conducting training and drills, prepositioning supplies, and establishing early warning systems.
- **Response:** Actions taken immediately before, during, and directly after a disaster to save lives, reduce health impacts, ensure public safety, and meet the basic subsistence needs of the people affected.
- **Recovery:** Actions taken after a disaster to restore and improve the pre-disaster living conditions of the affected community, while encouraging adjustments to reduce future disaster risk.

1.0.3 The Sendai Framework and the All-Hazards Approach

The preeminent global agreement guiding disaster risk management is the Sendai Framework for Disaster Risk Reduction 2015–2030. This framework marked a significant evolution in global policy by broadening the scope of disaster management to encompass an "all-hazards approach." This approach moves beyond a singular focus on natural events to include "man-made (now termed as human-induced) hazards and related biological, technological and environmental risks".[4]

The first priority of the Sendai Framework is "Understanding disaster risk," which underscores the global shift toward proactive risk reduction.[4] This philosophy recognizes that disasters are largely the result of unmanaged development, poverty, environmental degradation, and weak governance. Consequently, the most effective way to save lives and protect assets is not merely to respond more efficiently but to reduce vulnerability and enhance capacity *before* a hazard strikes.

1.0.4 Key Terminology in Disaster Risk Reduction

A clear understanding of core concepts is essential for implementing effective disaster risk reduction (DRR) strategies. Based on the UNDRR's globally accepted terminology, these concepts include [1]:

- **Hazard:** A process, phenomenon, or human activity that may cause loss of life, injury or other health impacts, property damage, social and economic disruption, or environmental degradation.
- **Exposure:** The situation of people, infrastructure, housing, production capacities, and other tangible human assets located in hazard-prone areas.
- **Vulnerability:** The conditions determined by physical, social, economic, and environmental factors or processes which increase the susceptibility of an individual, a community, assets, or systems to the impacts of hazards.
- **Capacity:** The combination of all the strengths, attributes, and resources available within an organization, community, or society to manage and reduce disaster risks and strengthen resilience.

Disasters can also be categorized by their characteristics, which influences the required preparedness and response strategies [1]:

- **Scale:**
  - **Small-scale disaster:** Affects local communities and requires assistance from beyond the immediate community.
  - **Large-scale disaster:** Affects a society to the extent that it requires national or international assistance.
- **Onset:**
  - **Sudden-onset disaster:** Triggered by a rapid or unexpected event, such as an earthquake, flash flood, or chemical explosion.
  - **Slow-onset disaster:** Emerges gradually over time, such as drought, sea-level rise, or desertification.
- **Frequency:** Disasters can be frequent or infrequent, depending on the probability of a given hazard. The cumulative impact of frequent, smaller-scale disasters can become chronic and severely undermine a community's resilience.

A Comprehensive Classification of Disasters

2.0 The UNDRR-ISC Hazard Classification Framework

Effective disaster risk reduction on a global scale is contingent upon a common, scientifically grounded language for identifying and describing hazards. The absence of a standardized lexicon impedes international cooperation, complicates data collection for risk assessment, and hinders the development of coherent policies.[4] To address this gap, the United Nations Office for Disaster Risk Reduction (UNDRR) and the International Science Council (ISC) have led a multi-stakeholder initiative to create a definitive classification of hazards.

2.0.1 The Hazard Information Profiles (HIPs)

This report adopts the UNDRR-ISC Hazard Information Profiles (HIPs) as its primary classification system. The HIPs represent a "trusted source of scientifically grounded, standardized hazard information," developed through a rigorous peer-review process involving over 330 authors and reviewers from more than 150 organizations, including UN agencies, academic institutions, and the private sector.[6] The 2025 revision of the HIPs, which defines and describes 282 distinct hazards, serves as the most comprehensive and authoritative global standard for hazard classification and is the foundation for the structure of this document.[6]

2.0.2 The Eight Primary Hazard Types

The UNDRR-ISC framework organizes the vast landscape of hazards into eight primary types. This categorization provides a logical structure for understanding risk and developing targeted management strategies. The eight types are: Hydrological and Meteorological, Extraterrestrial, Geological, Environmental, Chemical, Biological, Technological, and Societal.[6]

Hazard Type (Level 2)	Hazard Sub-Group (Illustrative Examples)	Core Definition	Key Characteristics

Geological	Earthquake, Volcanic Eruption, Tsunami, Landslide	Hazards originating from internal Earth processes.	Typically sudden-onset with potential for catastrophic, widespread physical damage.
Hydrological & Meteorological	Flood, Cyclone/Hurricane, Drought, Extreme Heat	Hazards of atmospheric, hydrological, or oceanographic origin.	Can be sudden-onset (flash floods, storms) or slow-onset (drought); often predictable to some degree; impacts can be regional or continental.
Environmental	Wildfire, Desertification, Biodiversity Loss	Hazards involving the degradation of the natural environment.	Often slow-onset and can be exacerbated by human activity and climate change; impacts are long-term and systemic.
Biological	Epidemic/Pandemic, Pest Infestation, Zoonotic Disease	Hazards of organic origin or conveyed by biological vectors.	Onset can be rapid or slow; potential for global-scale impact on public health and economies; containment is a primary challenge.
Chemical	Industrial Chemical Spill, Gas Leak, Contamination	Hazards originating from the release of chemical	Typically sudden-onset, localized events with severe health

		substances.	and environmental impacts; often linked to technological or industrial failures.
Technological	Industrial Accident, Structural Collapse, Power Outage	Hazards originating from the failure of human-made systems and infrastructure.	Generally sudden-onset; can be caused by human error, negligence, or system failure; often trigger cascading failures in other sectors.
Societal	Civil Unrest, Financial Crisis, Armed Conflict	Hazards originating from social, political, and economic actions and behaviors.	Onset can be sudden or gradual; driven by complex social dynamics; impacts are primarily on social cohesion, governance, and economic stability.
Extraterrestrial	Meteorite Impact, Space Weather (Solar Flare)	Hazards originating from asteroids, comets, or the Sun.	Low-probability, high-impact events; onset can be sudden (impact) or with some warning (space weather); potential for global-scale disruption.

While this framework categorizes hazards for analytical clarity, a modern understanding of disaster risk must account for their profound interconnectedness. A single initiating event frequently triggers a cascade of secondary disasters across different categories, creating complex, systemic risks that defy traditional, siloed response mechanisms. For example, a geological event like an earthquake can trigger a hydrological event (tsunami), further geological events (landslides), and technological failures (fires from ruptured gas lines).[9] Similarly, a meteorological hazard like a drought can exacerbate environmental hazards (wildfires) and societal hazards (conflict over scarce resources), leading to what is known as a Complex Humanitarian Emergency.[11] The recognition of this interconnectedness, formally explored by the ISC and UNDRR in work on "Systemic Risk," is crucial for developing the integrated, multi-sectoral preparedness and response strategies necessary to build true resilience in an increasingly complex world.[7]

Geological Hazards

Geological hazards originate from the Earth's internal processes. They are often characterized by their sudden onset and immense power, capable of causing catastrophic destruction within minutes. Understanding their mechanisms, preparing for their impact, and developing robust response and recovery protocols are fundamental to mitigating their devastating potential.

3.1 Earthquakes

An earthquake is a sudden, violent shaking of the ground, caused by the release of energy stored in the Earth's crust. These events are among the most destructive natural hazards, capable of leveling cities and triggering a cascade of secondary disasters.

3.1.1 Understanding the Hazard

Definition and Causes

An earthquake is caused by a sudden slip along a fault, which is a fracture between two blocks of rock.[14] The Earth's outer shell is composed of large, slowly moving puzzle pieces called tectonic plates. As these plates grind past one another, friction at their boundaries causes them to get stuck. Stress builds up over time until it overcomes the friction, causing a rapid release of energy in the form of seismic waves. These waves radiate outward from the earthquake's starting point (the hypocenter) and cause the ground shaking felt at the surface (the epicenter).[15]

Impacts and Effects

The primary effects of an earthquake are ground shaking, surface faulting (the visible rupture of the Earth's surface), and ground failure, including liquefaction, where saturated soil temporarily loses strength and behaves like a liquid.[17] These phenomena lead to the catastrophic failure of the built environment. Buildings, bridges, dams, roads, and utility lines (water, gas, electricity) can collapse or be severely damaged, causing the majority of casualties and economic losses.[17]

Earthquakes also frequently trigger dangerous secondary hazards. The displacement of the seafloor can generate massive tsunamis, intense ground shaking can cause devastating landslides and avalanches in mountainous regions, and ruptured gas and power lines can ignite widespread fires.[9]

Health Consequences

The health impacts of earthquakes are severe and occur in distinct phases. The immediate phase is dominated by trauma-related deaths and injuries from collapsing structures and secondary events like fires or tsunamis.[9] In the medium term, survivors face numerous health threats, including the secondary infection of untreated wounds, a heightened risk of communicable diseases (such as cholera or measles) in crowded and unsanitary temporary shelters, and interruptions in care for chronic conditions like diabetes or heart disease. The destruction of health facilities and transportation networks severely cripples the ability to provide medical care. Furthermore, the immense psychological trauma results in widespread and long-lasting psychosocial needs within the affected population.[9]

Monitoring and Detection

Earthquakes are detected and measured using seismometers, sensitive instruments that record ground vibrations.[20] A global network of seismograph stations continuously monitors seismic activity. When an earthquake occurs, it generates different types of seismic waves that travel at different speeds. The first to arrive are the compressional P-waves, followed by the shearing S-waves.[15] By measuring the time difference between the arrival of the P- and S-waves at a minimum of three separate stations, seismologists can triangulate the location of the earthquake's epicenter and calculate its magnitude on scales like the Richter or Moment Magnitude scale.[21]

3.1.2 User Preparedness

Individual and Family Preparedness

Personal preparedness is the cornerstone of earthquake safety. The recommended approach involves a four-step process to create a resilient household [23]:

1. **Secure Your Space:** Identify and mitigate home hazards. Heavy items like bookcases, refrigerators, and water heaters should be securely fastened to wall studs. Large and breakable objects should be moved to lower shelves. This simple step can prevent injuries from falling or flying objects during shaking.[10]
2. **Plan to Be Safe:** Develop a comprehensive family emergency plan. This plan should include an out-of-state contact person for communication, designated meeting places (both near the home and outside the neighborhood), and practiced evacuation routes.[23]
3. **Organize Disaster Supplies:** Assemble emergency kits for your home, car, and workplace. These kits should contain a minimum of three days' worth of non-perishable food and water, a first-aid kit, essential medications, a flashlight, a battery-powered or hand-crank radio, and tools like a wrench to turn off utilities.[23]
4. **Minimize Financial Hardship:** Organize important documents (identification, insurance policies, bank records) in a waterproof, portable container. Review your insurance policies; standard homeowner's insurance does not typically cover earthquake damage, requiring a separate policy.[23]

Community and Structural Preparedness

At the community level, the single most effective preparedness measure is the development, adoption, and rigorous enforcement of modern seismic building codes. These codes are designed to ensure that structures possess four key virtues: good structural configuration, adequate lateral strength and stiffness to resist horizontal forces, and, most importantly, good ductility—the ability to bend and deform without collapsing.[28] In seismically active nations like India, specific codes such as IS 1893 (Criteria for Earthquake Resistant Design) and IS 13920 (Ductile Detailing of Reinforced Concrete Structures) are mandatory in high-risk zones to ensure new construction is life-safe.[28]

Beyond building codes, community preparedness involves public education campaigns and regular, large-scale drills. Events like the annual Great ShakeOut provide millions of people with the opportunity to practice the "Drop, Cover, and Hold On" safety procedure, building muscle memory that is crucial during a real event.[19]

3.1.3 Response

Immediate Safety Procedure: Drop, Cover, and Hold On

The universally recommended and scientifically supported action to take the moment earthquake shaking begins is **Drop, Cover, and Hold On**.[24] This simple procedure is designed to protect individuals from falling debris and flying objects, which are the primary causes of injury and death during an earthquake.

- **DROP** to your hands and knees. This position prevents you from being knocked down and allows you to crawl to shelter.
- **COVER** your head and neck with one arm and hand. If a sturdy table or desk is nearby, crawl underneath it. If not, crawl next to an interior wall, away from windows.
- **HOLD ON** to your shelter (e.g., a table leg) until the shaking stops. Be prepared to move with it if the shaking shifts it.

Specific guidance exists for different scenarios. If in bed, stay there, turn face down, and cover your head with a pillow. If outdoors, move to an open area away from buildings, trees, and power lines. If driving, pull over to a clear location and remain in the vehicle with your seatbelt on. Individuals with mobility impairments should lock their wheels (if in a wheelchair), get as low as possible, and cover their head and neck.[30]

Search and Rescue (SAR) Operations

In the aftermath of a major earthquake with significant structural collapse, highly specialized Urban Search and Rescue (USAR) teams are deployed to locate and extricate trapped survivors. These teams use a combination of trained personnel and advanced technology [32]:

- **Canine Search Teams (Sniffer Dogs):** Dogs possess an extraordinary sense of smell and can cover large, complex rubble piles far more quickly than humans, detecting the scent of survivors buried deep within the debris.[33]
- **Electronic Search Equipment:**
  - **Seismic/Acoustic Sensors:** These highly sensitive listening devices can detect faint vibrations caused by survivors, such as tapping, scratching, or shouting.[32]
  - **Search Cameras:** Also known as "snake eyes," these are miniature cameras on flexible or rigid poles that can be inserted into small voids to visually search for victims and assess conditions.[32]
  - **UWB Rescue Radar:** This technology uses Ultra-Wide Band electromagnetic waves that can penetrate concrete and rubble to detect movement. It is sensitive enough to detect the subtle motion of breathing, making it possible to locate unconscious survivors.[32]
  - **Thermal Imaging Cameras:** These cameras detect heat signatures, allowing rescuers to identify victims in dark, dusty, or smoky environments by their body heat.[32]

National and International Response Mechanisms

The response to a major earthquake is a multi-layered effort involving local, national, and international actors.

- **National Response:** Many nations maintain specialized disaster response forces. India's **National Disaster Response Force (NDRF)** is a prime example. Formed under the Disaster Management Act of 2005, the NDRF is a dedicated force trained and equipped for complex rescue operations. Its expertise is globally recognized, having deployed to major international earthquakes, including the 2015 Nepal earthquake and the 2023 Türkiye earthquake, where its teams performed life-saving rescues.[34]
- **International Response:** When a disaster overwhelms a nation's capacity, international assistance is mobilized. Key international responders include:
  - **United Nations Children's Fund (UNICEF):** UNICEF maintains the world's largest humanitarian supply warehouse and can deliver life-saving supplies anywhere within 48 to 72 hours. Its response focuses on the unique needs of children, providing emergency health care, nutrition, safe water, sanitation, psychosocial support, and establishing child-friendly spaces.[36]

  - **The International Red Cross and Red Crescent Movement:** As the world's largest humanitarian network, the Red Cross provides a wide range of services. This includes distributing relief items (shelter kits, hygiene supplies), providing financial support, deploying disaster specialists (in logistics, communications, etc.), and operating its Restoring Family Links program to reconnect families separated by the crisis.[38]

3.1.4 Recovery

Post-Quake Safety Assessment

The period immediately following an earthquake is fraught with danger. The primary shock is often followed by a series of aftershocks, which can cause further damage to weakened structures. Individuals must be prepared to Drop, Cover, and Hold On repeatedly.[10]

Before moving, a careful assessment for hazards is critical. These include [10]:

- **Fires:** From broken gas lines or electrical shorts.
- **Gas Leaks:** Indicated by a distinct odor. If a leak is suspected, the main gas valve should be shut off immediately, and no open flames or electrical switches should be used.
- **Downed Power Lines:** Always assume they are live and maintain a safe distance.
- **Structural Damage:** Damaged buildings should be evacuated and not re-entered until deemed safe by professionals.

Health and Sanitation

Immediate first aid should be administered for injuries. In the days following, maintaining hygiene is crucial to prevent disease outbreaks. Food and water supplies must be checked for contamination from broken glass or ruptured pipes. Water from water heaters or melted ice cubes can be a safe source, but water from pools or spas should be avoided due to chemical content.[26] The disruption to healthcare services can be profound, making access to care for both new injuries and pre-existing conditions extremely difficult.[9]

Infrastructure and Rebuilding

Long-term recovery is a complex and lengthy process. In the United States, the **National Disaster Recovery Framework (NDRF)** provides a flexible structure to guide and coordinate recovery efforts among local, state, tribal, and federal partners. It focuses on restoring not just physical structures but also the social, economic, and environmental fabric of the community, with an emphasis on rebuilding stronger and more resilient than before.[40] This process involves detailed damage assessments, debris removal, restoration of essential services, and the long-term planning and reconstruction of homes, businesses, and public infrastructure.

Hydrological and Meteorological Hazards

Hydrological and meteorological hazards are those of atmospheric, hydrological, or oceanographic origin. They represent some of the most frequent and widespread disasters, driven by weather and climate systems. Their impacts range from the rapid, violent destruction of flash floods and cyclones to the slow, creeping devastation of drought. With climate change altering weather patterns globally, the frequency and intensity of these hazards are increasing, making preparedness and adaptation more critical than ever.

4.1 Floods

A flood is the overflowing of a large amount of water beyond its normal confines, especially over what is normally dry land.[42] It is one of the most common and costly natural disasters, affecting communities on every continent.

4.1.1 Understanding the Hazard

Definition and Types

Flooding is a multifaceted hazard with several distinct types, each with different causes and characteristics:

- **Fluvial (River) Floods:** This is the most common type of flooding, occurring when a river or stream overflows its banks. This is typically caused by prolonged, widespread rainfall, rapid snowmelt, or an ice jam blocking the river's flow. The rise in water level can be gradual, providing some warning time.[43]
- **Pluvial (Surface Water) Floods:** These floods occur when intense rainfall overwhelms the capacity of natural and man-made drainage systems. The excess water flows overland and ponds in low-lying areas. This category includes **flash floods**, which are characterized by a rapid and extreme flow of high water. They are often caused by intense thunderstorms and are particularly dangerous due to their sudden onset and destructive power.[42]
- **Coastal Floods:** This type of flooding occurs in coastal areas when seawater inundates the land. It is primarily caused by **storm surge**—an abnormal rise in sea level generated by a severe storm's winds and low atmospheric pressure—or by tsunamis. Coastal flooding is exacerbated when storm surge coincides with a high tide.[44]
- **Groundwater Floods:** This less common type occurs when the level of water stored in the ground (the water table) rises to the surface as a result of prolonged, heavy rainfall. While the water rises slowly, this type of flooding can persist for extended periods.[45]

Impacts and Effects

The impacts of flooding are extensive and devastating. The primary and most immediate impact is the loss of human life, predominantly through drowning.[46] The force of moving water can destroy buildings, wash away bridges, and erode roads and foundations. Widespread property damage is a hallmark of major flood events, affecting homes, businesses, and public infrastructure.[46] In agricultural regions, floods can destroy entire harvests and lead to the loss of livestock, causing severe economic hardship and threatening food security.[47] The disruption to transportation, power, and communication networks can paralyze entire regions, with economic impacts extending far beyond the flooded area.[46]

Health Consequences

Floodwater poses a severe threat to public health. It is often a toxic mixture of raw sewage, agricultural and industrial chemicals, debris, and pathogens.[46] Direct contact can lead to skin rashes, wound infections, and gastrointestinal illnesses.[48] The contamination of drinking water supplies is a major concern, often necessitating boil-water advisories to prevent outbreaks of diseases like cholera and dysentery.[49]

After the floodwaters recede, the health risks continue. Standing water creates ideal breeding grounds for mosquitoes, increasing the risk of vector-borne diseases like Dengue and Zika virus.[48] The most persistent long-term health hazard is mold. Saturated building materials provide a perfect environment for mold to grow, which can cause severe respiratory problems, particularly for individuals with asthma or compromised immune systems.[50] Beyond the physical ailments, the trauma of experiencing a flood, losing a home, and facing a lengthy recovery process can lead to significant and lasting mental health issues, including anxiety, depression, and post-traumatic stress disorder (PTSD).[46]

Monitoring and Forecasting

Modern flood forecasting and early warning systems are crucial for mitigating the impact of floods. These systems rely on a combination of real-time monitoring and advanced predictive modeling.[51]

- **Monitoring:** A network of sensors is used to track environmental conditions. These include staff gages (large rulers), submersible pressure transducers, and bubblers to measure river levels, as well as non-contact radar and ultrasonic sensors that measure water height from above.[52]
- **Forecasting:** The data from these sensors, along with meteorological forecasts for rainfall, are fed into sophisticated computer models. These include hydrological models, which forecast the amount of water flowing in a river, and inundation models, which predict which specific areas will be flooded and to what depth. Increasingly, Artificial Intelligence (AI) is being used to improve the accuracy and lead time of these forecasts, with some systems now able to provide reliable warnings up to seven days in advance.[51] These warnings are then disseminated to the public through multiple channels, including emergency alerts, media broadcasts, and sirens.

4.1.2 User Preparedness

Individual and Family Preparedness

Effective flood preparedness at the household level can significantly reduce risk and improve safety. Key steps include [49]:

1. **Know Your Risk:** The first step is to understand your vulnerability. Individuals should use official resources, such as the Federal Emergency Management Agency's (FEMA) Flood Map Service Center in the U.S., to determine if their property is located in a flood-prone area.[49]
2. **Make a Plan:** Every family should have a disaster plan that includes a communication strategy and pre-determined evacuation routes to higher ground. This plan should be practiced regularly.[54]
3. **Build an Emergency Kit:** An emergency kit should be prepared with essential supplies, including several days' worth of food and water, a first-aid kit, medications, a flashlight, a radio, and important documents in a waterproof container.[49]
4. **Protect Your Home:** In high-risk areas, proactive measures can reduce damage. This includes elevating critical utilities like the furnace, water heater, and electrical panel; installing "check valves" in sewer lines to prevent floodwater from backing up; and sealing basement walls with waterproofing compounds.[54]

Community Preparedness

Resilient communities are built on a foundation of collective preparedness. This involves robust public awareness campaigns to educate residents about flood risks and safety measures.[55] Establishing reliable, community-wide warning systems and clearly marked evacuation routes and shelters is essential. Programs like the U.S. National Flood Insurance Program's (NFIP) Community Rating System (CRS) provide a valuable framework, offering discounts on flood insurance premiums to communities that adopt and enforce higher standards for flood risk reduction, such as preserving open space in floodplains and enforcing stricter building codes.[57]

4.1.3 Response

Immediate Safety Procedure: "Turn Around, Don't Drown!"

The single most important safety rule during a flood is to **never enter floodwaters**. The slogan "Turn Around, Don't Drown!" encapsulates this critical advice.[49] The dangers are often underestimated:

- Just 6 inches of fast-moving water can knock an adult off their feet.[49]
- As little as 12 to 24 inches of moving water can sweep away most vehicles, including SUVs and trucks.[49]
- The water may hide washed-out sections of road, debris, or downed power lines, posing hidden dangers.[56]

Evacuation

When authorities issue an evacuation order, it must be heeded immediately. If time allows before leaving, utilities should be turned off at the main switches, and essential items and valuables should be moved to the highest possible floor.[54]

Rescue Operations and Equipment

Flood rescue is a specialized discipline requiring trained personnel and specific equipment. Teams like India's NDRF are frequently deployed for large-scale flood rescue operations.[60] Key equipment includes:

- **Watercraft:** Inflatable Rescue Boats (IRBs) and more rigid Fiberglass Reinforced Plastic (FRP) boats, typically equipped with powerful outboard motors, are the primary vehicles for accessing flooded areas and evacuating people.[63]
- **Personal Protective Equipment (PPE):** Rescuers wear specialized gear for protection, including waterproof dry suits or wetsuits, helmets, and high-buoyancy personal flotation devices (PFDs), commonly known as life jackets. Life jackets are also provided to all survivors.[63]
- **Advanced Technology:** Drones have become an invaluable tool in flood response. They are used for aerial reconnaissance to assess the extent of flooding, search for stranded individuals using thermal imaging, and in some cases, deliver life-saving equipment like flotation devices directly to people in the water.[64]
- **Diving and Underwater Gear:** For scenarios involving submerged vehicles or structures, rescue teams are equipped with personnel diving kits and underwater communication

sets to conduct searches and rescues below the surface.[63]

National and International Response Mechanisms

In India, the **National Disaster Management Authority (NDMA)** is the apex body responsible for creating national flood management policies and guidelines. This includes commissioning technical studies, producing Flood Hazard Atlases for vulnerable states, and developing specific strategies for mitigating the growing threat of urban flooding.[66]

Internationally, organizations like the **American Red Cross** play a vital role in the immediate response, setting up shelters to provide safe lodging, food, and comfort to displaced families.[55]

UNICEF's response prioritizes the needs of children, focusing on providing safe drinking water, hygiene kits to prevent disease, and establishing Child-Friendly Spaces where children can play, learn, and receive psychosocial support in a safe environment.[71]

4.1.4 Recovery

The recovery phase after a flood is often a long, arduous, and expensive process. It requires a systematic approach to ensure safety and prevent long-term health problems.

Returning Home Safely

Residents should not return to their homes until authorities have declared the area safe. The initial return is a critical period for safety assessment.[72] Key checks include looking for any structural damage to the foundation or walls, and ensuring that electricity and gas are safely shut off at the main source before entering. The risk of electrocution is high, and a qualified electrician should inspect the system before power is restored.[49] The ground will be littered with hidden hazards like broken glass and nails, and mud-covered surfaces are extremely slippery.[49]

Health and Sanitation

Protecting health during cleanup is paramount. All food—including canned goods—that has come into contact with floodwater should be discarded as it may be contaminated.[49] Tap water should not be consumed until local authorities have confirmed its safety; until then, it should be boiled for at least one minute before use.[49] Personal protective equipment, including waterproof boots, gloves, and N95-rated masks, is essential during cleanup to protect against contaminants and mold spores.[49]

Cleaning and Repairing a Flooded Home

The process of remediating a flooded home is meticulous and must be done correctly to ensure the home is safe and habitable [50]:

1. **Document Damage:** Before any cleanup begins, homeowners should thoroughly document all damage with photographs and videos for their insurance claim. Samples of damaged items like carpet and upholstery should be kept.[72]
2. **Dry Out the Home:** The most urgent task is to dry the building and its contents as quickly as possible. The goal is to dry everything within 24-48 hours to prevent the growth of mold. This involves removing all wet items, opening all doors and windows for ventilation, and using fans, heaters, and dehumidifiers to accelerate the process.[50]
3. **Clean and Disinfect:** All hard surfaces that came into contact with floodwater must be washed with soap and water and then disinfected, typically with a solution of household bleach and water. This is especially critical for kitchen surfaces and utensils.[50]
4. **Address Mold Growth:** If mold is visible, it must be cleaned immediately. Small areas can be cleaned with a bleach solution while wearing a P2/N95 mask. For extensive mold contamination, a professional remediation specialist should be hired.[50]
5. **Discard Porous Materials:** Many items that absorb water cannot be safely cleaned and must be discarded. This includes mattresses, upholstered furniture, carpets and padding, and drywall that has been saturated.[50]

Technological Hazards

Technological hazards are those that originate from the failure of human-made systems, infrastructure, and technologies. These disasters are often characterized by their sudden onset and can result from human error, negligence, or systemic design flaws. They range from industrial accidents and structural collapses to widespread power outages and cyberattacks. As societies become increasingly reliant on complex, interconnected technologies, the potential for cascading failures and large-scale disruption from these hazards grows.

6.1 Cyberattacks

A cyberattack is a malicious and deliberate attempt by an individual or organization to breach the information system of another individual or organization.[74] The goal is typically to steal, alter, destroy, or gain unauthorized access to computer systems, networks, and data.[75]

6.1.1 Understanding the Hazard

Definition and Types

Cyberattacks employ a wide variety of methods to exploit vulnerabilities in digital systems. The most common types include:

- **Malware:** An umbrella term for any malicious software designed to harm or exploit a computer or network. This includes viruses, which attach to clean files and spread; worms, which are self-replicating and can spread across networks without human interaction; Trojans, which disguise themselves as legitimate software; and spyware, which secretly gathers user information.[76]
- **Ransomware:** A particularly damaging form of malware that encrypts a victim's files or entire system, rendering them inaccessible. The attacker then demands a ransom payment, typically in cryptocurrency, in exchange for the decryption key. Ransomware is one of the most prevalent and costly forms of cybercrime.[76]
- **Phishing:** This is a form of social engineering where attackers use fraudulent emails, text messages, or websites designed to look like they are from a legitimate source. The goal is to trick victims into revealing sensitive information, such as passwords and credit card numbers, or to deploy malware onto their device.[76]
- **Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) Attacks:** These attacks aim to make a machine or network resource unavailable to its intended users. This is achieved by flooding the target with a massive volume of traffic or requests,

overwhelming its capacity and causing it to crash or become unresponsive. A DDoS attack uses multiple compromised computer systems (a "botnet") to launch the attack, making it more powerful and harder to trace.[75]

Impacts and Effects

The impact of cyberattacks is vast and multifaceted. For individuals, the consequences can include identity theft, financial loss through fraud, and extortion.[80] For businesses, a successful attack can lead to devastating financial losses from theft, business disruption, and reputational damage. The global economic cost of cybercrime is staggering, projected to reach $10.5 trillion per year by 2025.[80]

The most significant strategic risk arises from cyberattacks targeting **critical infrastructure**. These are the assets, systems, and networks so vital to a nation that their incapacitation or destruction would have a debilitating effect on security, economic stability, and public health or safety.[81] Modern power grids, water treatment plants, transportation systems, financial networks, and healthcare facilities are all heavily reliant on industrial control systems (ICS) and SCADA systems, which are increasingly targeted by sophisticated actors, including nation-states.[81] A successful attack could trigger a power outage affecting millions, contaminate a city's water supply, or shut down a hospital, causing physical damage and potential loss of life.[82]

6.1.2 User Preparedness

Individual Cybersecurity Hygiene

Just as personal hygiene prevents the spread of disease, good cybersecurity hygiene can prevent the vast majority of cyberattacks. Essential practices for individuals include [83]:

- **Strong Password Practices:** Use long, complex, and unique passwords for every account. A password manager is a critical tool for creating and storing these credentials securely.
- **Multi-Factor Authentication (MFA):** Enable MFA (also known as two-step verification) wherever it is offered. This provides a crucial second layer of security beyond just a

password.

- **Software Updates:** Keep all software, including operating systems and applications, up-to-date. These updates frequently contain patches for security vulnerabilities that attackers could otherwise exploit.
- **Phishing Awareness:** Be skeptical of unsolicited emails, texts, or calls. Do not click on suspicious links or download unexpected attachments. Always verify the sender's identity through a separate, trusted channel.
- **Regular Backups:** Regularly back up important data to an external hard drive or a secure cloud service. This is the most effective defense against ransomware, as it allows for data restoration without paying the ransom.

Organizational Preparedness

Organizations must adopt a defense-in-depth strategy that assumes a breach is not a matter of *if*, but *when*. Key preparedness measures include [83]:

- **Technical Controls:** Implement a layered security architecture that includes network firewalls, endpoint protection software (antivirus/anti-malware), email filtering, and secure Wi-Fi configurations.
- **Access Control:** Enforce the principle of "least privilege," ensuring that employees only have access to the data and systems absolutely necessary for their jobs.
- **Vulnerability Management:** Regularly scan systems and networks for vulnerabilities and apply patches in a timely manner.
- **Employee Training:** Conduct regular, mandatory cybersecurity awareness training for all employees. This should include simulated phishing attacks to test and reinforce learning. A well-trained workforce is a critical line of defense.
- **Incident Response Plan:** Develop and regularly test a formal Incident Response Plan (IRP) that details the steps to be taken in the event of a breach.

National and International Agencies

Recognizing the systemic threat, nations have established specialized agencies to lead national cybersecurity efforts. In the United States, the **Cybersecurity and Infrastructure Security Agency (CISA)** leads the effort to protect federal networks and national critical infrastructure, while the **National Security Agency (NSA)** provides cybersecurity expertise and foreign intelligence.[88] In India, the

Indian Computer Emergency Response Team (CERT-In) is the national agency for responding to incidents, while the **National Critical Information Infrastructure Protection Centre (NCIIPC)** focuses on protecting vital sectors.[90] The

European Union Agency for Cybersecurity (ENISA) serves a similar coordinating role across Europe.[93] These agencies are responsible for issuing threat alerts, sharing intelligence, and coordinating national defense and response strategies.

6.1.3 Response and Recovery

Detection and Prevention Technologies

Organizations deploy a suite of technologies to monitor their digital environments for threats. **Intrusion Detection Systems (IDS)** and **Intrusion Prevention Systems (IPS)** analyze network traffic for suspicious patterns. **Security Information and Event Management (SIEM)** systems aggregate and correlate log data from across the network to identify potential security events. Modern **Endpoint Detection and Response (EDR)** solutions provide deep visibility into activity on individual computers and servers, using behavioral analysis to detect advanced threats.[83]

The Incident Response Plan (IRP)

When a security incident is detected, a swift and structured response is critical to minimize damage. An effective IRP, guided by established frameworks from bodies like the National Institute of Standards and Technology (NIST), typically follows six phases [78]:

1. **Preparation:** This ongoing phase involves establishing the policies, procedures, tools, and the Computer Security Incident Response Team (CSIRT) needed to handle an incident.
2. **Identification (or Detection & Analysis):** This phase involves detecting a potential security incident, analyzing all available data to confirm its validity, determining its scope, and prioritizing the response based on its potential impact.
3. **Containment:** The immediate goal is to prevent the incident from spreading further. This may involve isolating affected systems from the network, blocking malicious IP addresses, or disabling compromised user accounts.
4. **Eradication:** Once contained, the root cause of the incident is identified and removed

from the environment. This could involve removing malware, patching vulnerabilities, and resetting compromised credentials.
5. **Recovery:** Systems, data, and services are carefully restored to normal operation. This is typically done using clean backups that predate the incident. Systems are monitored closely to ensure the threat has been fully eliminated.
6. **Lessons Learned (or Post-Incident Review):** After every incident, a thorough review is conducted to understand what happened, how the response was handled, and what could be done to improve security and response processes in the future.

Individual Recovery

If an individual suspects they have been the victim of a cyberattack or identity theft, they should act immediately to limit the damage [84]:

- **Change Passwords:** Immediately change the passwords for any compromised accounts and any other accounts that use the same or similar passwords.
- **Contact Financial Institutions:** Notify banks and credit card companies of any potential fraud. They can freeze accounts or issue new cards.
- **Report the Incident:** File a report with the appropriate authorities, such as the FBI's Internet Crime Complaint Center (IC3) and local police. This creates an official record of the event.
- **Place Fraud Alerts:** Contact the major credit bureaus (Equifax, Experian, TransUnion) to place a fraud alert or credit freeze on your file, making it harder for criminals to open new accounts in your name.

Societal Hazards

Societal hazards are those that originate from the actions, behaviors, and structures of human societies. Unlike hazards rooted in natural or technological processes, these events are driven by complex social, political, and economic dynamics. They include phenomena such as civil unrest, armed conflict, and financial crises. These hazards can be both a cause and a consequence of other disasters, creating feedback loops that destabilize communities and nations.

7.1 Civil Unrest (Protests, Riots, Civil Disorder)

Civil unrest encompasses a range of collective actions by groups of people that disrupt public order. These events are often a manifestation of deep-seated societal grievances and can have profound and lasting impacts on a community's social fabric, economy, and political stability.

7.1.1 Understanding the Hazard

Definition and Types

The UNDRR defines civil unrest as "sporadic but continued collective physical violence in a context of social or political instability".[99] This hazard exists on a spectrum of intensity [100]:

- **Non-Violent Forms:** These include lawful and peaceful actions such as organized demonstrations, marches, strikes, and protests. While generally non-violent, these gatherings can disrupt normal activities and sometimes escalate.
- **Disruptive Forms (Civil Disorder):** This involves actions that, while not necessarily violent, directly interfere with public order, such as blocking roadways or buildings.
- **Violent Forms (Riots):** A riot is a violent public disturbance where a crowd becomes a mob, engaging in destructive and harmful acts such as vandalism, arson, looting, and assault on individuals, including law enforcement. Riots can be further categorized based on their primary motivation, such as communal (between ethnic or religious groups), protest, commodity (e.g., food riots), or celebration riots (e.g., after a sports event).[102]

Causes and Triggers

Civil unrest is rarely caused by a single event but rather emerges from a complex interplay of underlying grievances and precipitating triggers. The primary drivers include [103]:

- **Economic Inequality:** Significant gaps between the rich and poor, high unemployment, inflation, and economic hardship are powerful motivators for unrest.
- **Political Oppression and Corruption:** A lack of political rights, perceived fraudulent elections, government corruption, and repressive state actions erode public trust and can lead to mass protests demanding accountability and reform.

- **Social Injustice and Identity Politics:** Grievances related to systemic discrimination based on race, ethnicity, religion, or other identity markers are a frequent cause. Tensions can be deliberately fueled by political actors who use scapegoating and misinformation to mobilize support.[103]
- **Environmental Factors:** Climate change is an emerging and accelerating driver of unrest. Resource scarcity (e.g., water or food shortages), displacement due to extreme weather events, and dissatisfaction with government environmental policies can all trigger or exacerbate conflict.[103]

Impacts and Effects

The consequences of civil unrest are severe and far-reaching. Violent events directly result in deaths, injuries, and widespread destruction of public and private property.[100] The economic impacts are profound, including business interruption from closures and supply chain disruptions, damage to infrastructure, capital flight as investors lose confidence, and a long-term decline in GDP.[108]

The social impacts are equally damaging. Unrest erodes social cohesion and trust between different community groups and between the public and authorities. It can disrupt essential services like education and healthcare. For individuals directly affected, the experience can lead to lasting psychological trauma, including anxiety and PTSD. In the long term, sustained unrest can lead to a "brain drain" as skilled individuals and families flee to safer, more stable environments, further hindering a community's recovery.[108]

Role of Social Media

Social media platforms play a powerful and dual-edged role in modern civil unrest. For activists and protesters, they are invaluable tools for organizing, mobilizing participants, and rapidly disseminating information and videos, often bypassing state-controlled media.[111] However, these same platforms are also exploited to spread disinformation, hate speech, and propaganda that can inflame tensions and incite violence.[114] Furthermore, governments and law enforcement agencies increasingly use social media as a surveillance tool to monitor activists and gather intelligence, raising significant privacy and human rights concerns.[112]

7.1.2 User Preparedness and Response

Individual and Community Preparedness

The primary preparedness strategy for individuals and communities is **situational awareness**. This involves staying informed about local political and social tensions, monitoring trusted media sources for information about planned demonstrations, and avoiding areas where unrest is occurring or likely to occur.[116] Individuals should have contingency plans for transportation, communication, and essential supplies in case unrest disrupts normal services. Businesses in vulnerable areas should develop continuity of operations plans to protect employees and assets.[100]

Safety Measures During Unrest

If one is unexpectedly caught in or near a protest or riot, personal safety is the immediate priority. Key safety measures include [116]:

- **Avoid and Evacuate:** The best course of action is to leave the area immediately. Move calmly to the edge of the crowd, looking for an escape route like a side street, alley, or safe building. Do not run, as this can draw unwanted attention.
- **Maintain a Low Profile:** Avoid confrontation or engagement with protesters or law enforcement. Do not wear clothing or symbols associated with any particular group.
- **Vehicle Safety:** If driving, turn away from the area. If the vehicle is surrounded and cannot move, it may be safer to park, lock the vehicle, and leave on foot to seek shelter. Do not attempt to drive through a crowd.
- **Prepare for Crowd Control Agents:** Be aware that law enforcement may use tear gas or other chemical agents. It is advisable not to wear contact lenses to protests. If exposed, move upwind and away from the source of the gas.

De-escalation Techniques

De-escalation is a vital skill for preventing tense situations from turning violent. These techniques can be used by law enforcement, community leaders, and individuals to reduce emotional intensity. Core strategies include [121]:

- **Control Your Own Response:** Remain calm and composed. Use controlled breathing to manage your own stress. Maintain neutral, non-threatening body language (e.g., open hands, relaxed posture).
- **Effective Communication:** Speak in a calm, soft, and even tone. Use simple, clear, and non-provocative language.
- **Active Listening and Empathy:** Give the agitated person space to speak. Acknowledge and validate their feelings without judgment, using phrases like, "I hear your frustration," or "I can see why you are upset." This demonstrates that you are listening and can help lower their defenses.

Role of Law Enforcement

The role of law enforcement during civil unrest is to maintain public order and ensure safety while upholding the constitutional right to peaceful assembly. This requires a delicate balance and a well-defined strategy. Effective management involves robust pre-event planning, clear communication protocols, and strong inter-agency coordination, often through a **unified command** structure.[124] Law enforcement must also have plans to protect critical infrastructure, manage mass arrests if necessary, and engage with the media. In the wake of major events like the January 6, 2021, attack on the U.S. Capitol, there has been a significant increase in emphasis on joint training exercises among local, state, and federal agencies to improve coordination and response capabilities for future events.[126]

7.1.3 Recovery and Reconciliation

The aftermath of significant civil unrest requires more than just physical cleanup; it necessitates a long-term process of social and political healing.

Post-Conflict Reconstruction

Recovery is a multidimensional process aimed at rebuilding a society's physical, social, economic, and political infrastructure. It involves restoring security, re-establishing essential services, revitalizing the economy, and addressing the deep-seated grievances that led to the conflict in the first place.[128]

Community-Led Peacebuilding

Evidence suggests that the most sustainable peace is built from the ground up. **Community-based approaches** empower local organizations, civil society groups (CSOs), and community leaders to take ownership of the peacebuilding process. These grassroots initiatives are better positioned to understand local conflict dynamics and design culturally appropriate solutions. They focus on facilitating dialogue, mediating disputes, and rebuilding trust and social relationships within and between communities.[130]

The Reconciliation Process

Reconciliation is the deep, long-term process by which a society moves from a divided past to a shared future. It is not a quick fix but involves fundamental changes in attitudes, emotions, and relationships. While the path can vary, essential components of a successful reconciliation process include [134]:

- **Justice:** Holding perpetrators of violence accountable through formal or traditional justice mechanisms.
- **Truth-Telling:** Creating platforms for victims and perpetrators to share their experiences, fostering a shared understanding of the past.
- **Reparations:** Providing redress to victims for the harm they have suffered.
- **Empathy:** Fostering a willingness among all parties to listen to and acknowledge the suffering of others.

Leaving these hurts and injustices unaddressed allows them to fester, undermining political and economic reforms and increasing the likelihood of future violence.[134]

Case Studies of Civil Unrest in India

India's history provides numerous case studies illustrating the complex dynamics of civil unrest, particularly communal violence.

- The **2020 Delhi Riots** were a wave of violence, primarily between Hindu and Muslim

groups, triggered by protests against the Citizenship Amendment Act. The riots resulted in 53 deaths, hundreds of injuries, and widespread property destruction, disproportionately affecting the Muslim community.[135]
- The **1983 Nellie Massacre** in Assam saw the killing of nearly 2,000 Muslims of Bengali origin by their neighbors amid tensions over immigration and electoral rolls, highlighting how political and ethnic grievances can erupt into extreme violence.[136]
- The ongoing ethnic conflict in **Manipur**, which began in 2023 between the Meitei and Kuki-Zo communities, demonstrates the explosive potential of long-standing disputes over land, resources, and political representation, resulting in over 140 deaths and the displacement of more than 54,000 people.[137]
- The targeting of religious sites has been a recurring trigger for major unrest. The **demolition of the Babri Masjid in 1992** led to nationwide riots that killed over 2,000 people. More recent events, such as the dispute over the **Gyanvapi Mosque** and the violence during a survey of the **Shahi Jama Masjid in Sambhal in 2024**, show that such tensions remain a potent source of conflict.[138] These cases often reveal the instrumental role political parties and partisan rhetoric can play in either inciting or mitigating violence.[140]

Conclusion

This report has provided a comprehensive examination of the multifaceted nature of disasters, structured according to the globally recognized all-hazards approach. The analysis underscores a critical evolution in the global perspective on disaster management: a decisive shift from a model of reactive response to one of proactive disaster risk reduction. The foundational understanding, as defined by the United Nations, is that disasters are not merely natural or technological events but are the result of the interaction between a hazard and a vulnerable society. This places the focus squarely on mitigating the underlying drivers of risk—poverty, poor governance, environmental degradation, and inadequate infrastructure—before a hazard can manifest as a catastrophe.

The adoption of a standardized, scientifically grounded classification of hazards, such as the UNDRR-ISC Hazard Information Profiles, is not an academic exercise but a practical necessity. It enables coherent international cooperation, facilitates accurate risk assessment, and allows for the development of targeted preparedness strategies. However, while categorization is essential for analysis, the evidence increasingly points to the interconnectedness of hazards and the rise of systemic risk. A single event can trigger a cascade of failures across geological, meteorological, technological, and societal domains, demanding integrated, multi-sectoral response strategies that transcend traditional institutional silos.

Across all hazard types—from the sudden violence of an earthquake to the insidious spread of a cyberattack or the complex eruption of civil unrest—a consistent theme emerges: the paramount importance of preparedness at every level of society.

- For **individuals and families**, preparedness is rooted in awareness, planning, and action. This includes understanding local risks, creating emergency plans and communication strategies, assembling supply kits, and practicing safety procedures like "Drop, Cover, and Hold On" for earthquakes or "Turn Around, Don't Drown!" for floods.
- For **communities**, resilience is built through the adoption and enforcement of strong building codes, the establishment of robust early warning systems, the protection of critical infrastructure, and the fostering of social cohesion through public education and community-led peacebuilding initiatives.
- For **institutions**, effective disaster management requires clear mandates, inter-agency coordination, specialized and well-equipped response forces like the NDRF in India, and participation in global humanitarian networks such as the Red Cross/Red Crescent Movement and UN agencies.

Ultimately, the capacity to withstand, adapt to, and recover from the diverse and growing array of threats facing the global community depends on a shared commitment to this culture of preparedness. By understanding the risks we face, investing in mitigation, and empowering individuals and communities with the knowledge and resources to act, we can move from a posture of vulnerability to one of resilience, building a safer and more secure future for all.

Works cited

1. Definition: Disaster - UNDRR, accessed on September 12, 2025, https://www.undrr.org/terminology/disaster
2. Disaster Services - American Red Cross, accessed on September 12, 2025, https://www.redcross.org/about-us/careers/pbjob-types/disaster-services.html
3. Disaster Preparedness | Disaster Relief Training - American Red Cross, accessed on September 12, 2025, https://www.redcross.org/take-a-class/disaster-training
4. UNDRR / ISC Sendai hazard definition and classification review - Understanding Risk, accessed on September 12, 2025, https://understandrisk.org/undrr-isc-sendai-hazard-definition-and-classification-review/
5. Hazard definition and classification review - UNECE, accessed on September 12, 2025, https://unece.org/sites/default/files/2023-01/S7_3_UNDRR_0.pdf
6. Hazard definition and classification review: Technical report (2025 ..., accessed on September 12, 2025, https://www.undrr.org/publication/documents-and-publications/hazard-definition-and-classification-review-technical-report
7. Review of hazards definition and classification - International Science Council, accessed on September 12, 2025, https://council.science/our-work/hazards-definition-and-classification/
8. Hazard Definition & Classification Review - UNDRR, accessed on September 12,

2025, https://www.undrr.org/media/107380/download
9. Earthquakes - World Health Organization (WHO), accessed on September 12, 2025, https://www.who.int/health-topics/earthquakes
10. Earthquakes | Ready.gov, accessed on September 12, 2025, https://www.ready.gov/earthquakes
11. Understanding Mortality Patterns in Complex Humanitarian Emergencies - NCBI, accessed on September 12, 2025, https://www.ncbi.nlm.nih.gov/books/NBK223340/
12. Complex Humanitarian Emergencies - Center for Disaster Philanthropy, accessed on September 12, 2025, https://disasterphilanthropy.org/resources/complex-humanitarian-emergencies/
13. Complex humanitarian emergency (CHE) | Description, Examples, & Aid - Britannica, accessed on September 12, 2025, https://www.britannica.com/topic/complex-humanitarian-emergency
14. www.mtu.edu, accessed on September 12, 2025, https://www.mtu.edu/geo/community/seismology/learn/earthquake-cause/#:~:text=Earthquakes%20are%20usually%20caused%20when,that%20make%20the%20ground%20shake.
15. The Science of Earthquakes | U.S. Geological Survey, accessed on September 12, 2025, https://www.usgs.gov/programs/earthquake-hazards/science-earthquakes
16. What is an earthquake and what causes them to happen? - USGS.gov, accessed on September 12, 2025, https://www.usgs.gov/faqs/what-earthquake-and-what-causes-them-happen
17. What are the Effects of Earthquakes? | U.S. Geological Survey, accessed on September 12, 2025, https://www.usgs.gov/programs/earthquake-hazards/what-are-effects-earthquakes
18. uwiseismic.com, accessed on September 12, 2025, https://uwiseismic.com/earthquakes/effects/#:~:text=Infrastructural%20Damage,heavy%20objects%20can%20be%20fatal.
19. Earthquakes and Faults | Department of Natural Resources - dnr.wa.gov, accessed on September 12, 2025, https://dnr.wa.gov/washington-geological-survey/geologic-hazards-and-environment/earthquakes-and-faults
20. www.bgs.ac.uk, accessed on September 12, 2025, https://www.bgs.ac.uk/discovering-geology/earth-hazards/earthquakes/how-are-earthquakes-detected/#:~:text=Seismometers%20allow%20us%20to%20detect,big%20a%20particular%20earthquake%20is.
21. How do seismologists locate an earthquake? | U.S. Geological Survey - USGS.gov, accessed on September 12, 2025, https://www.usgs.gov/faqs/how-do-seismologists-locate-earthquake
22. Seismographs - Keeping Track of Earthquakes | U.S. Geological Survey - USGS.gov, accessed on September 12, 2025, https://www.usgs.gov/programs/earthquake-hazards/seismographs-keeping-track-earthquakes

23. What can I do to be prepared for an earthquake? - USGS.gov, accessed on September 12, 2025, https://www.usgs.gov/faqs/what-can-i-do-be-prepared-earthquake
24. Earthquake Safety Tips | Mass.gov, accessed on September 12, 2025, https://www.mass.gov/info-details/earthquake-safety-tips
25. How to Prepare for an Earthquake - Tips for Your Preparedness Plan | CEA, accessed on September 12, 2025, https://www.earthquakeauthority.com/california-earthquake-risk/personal-preparedness
26. What do I do AFTER an earthquake? | U.S. Geological Survey, accessed on September 12, 2025, https://www.usgs.gov/faqs/what-do-i-do-after-earthquake
27. Earthquake Preparedness - Ventura County Fire Department, accessed on September 12, 2025, https://vcfd.org/earthquake-preparedness/
28. Earthquake Tip - BMTPC, accessed on September 12, 2025, https://www.bmtpc.org/DataFiles/CMS/file/EQTIPS_PDFFiles/EQTip11.pdf
29. www.shakeout.org, accessed on September 12, 2025, https://www.shakeout.org/centralus/downloads/shakeout_individual_drill_manual.pdf
30. Earthquake Safety - American Red Cross, accessed on September 12, 2025, https://www.redcross.org/get-help/how-to-prepare-for-emergencies/types-of-emergencies/earthquake.html
31. Drop, Cover and Hold - Get Ready (NZ, accessed on September 12, 2025, https://getready.govt.nz/emergency/earthquakes/drop-cover-hold
32. How Search and Rescue Teams find survivors after earthquakes?, accessed on September 12, 2025, https://www.fireproductsearch.com/how-search-and-rescue-teams-find-survivors-after-earthquakes/
33. Professional rescue team and rescue equipment after wenchuan earthquake - Disaster Risk Reduction Knowledge Service, accessed on September 12, 2025, https://ikcest-drr.data.ac.cn/tutorial/p98de
34. Making India More Disaster-Resilient - Press Release: Press Information Bureau, accessed on September 12, 2025, https://www.pib.gov.in/PressReleseDetailm.aspx?PRID=2113875
35. Home | NDRF - National Disaster Response Force, accessed on September 12, 2025, https://www.ndrf.gov.in/
36. Earthquake Relief | UNICEF USA, accessed on September 12, 2025, https://www.unicefusa.org/what-unicef-does/emergency-response/earthquakes
37. UNICEF Afghanistan Earthquake Response: the First Seven Days, accessed on September 12, 2025, https://www.unicefusa.org/stories/unicef-afghanistan-earthquake-response-first-seven-days
38. International Disaster Relief Efforts | American Red Cross, accessed on September 12, 2025, https://www.redcross.org/about-us/our-work/international-services/international-disasters-and-crises.html

39. Earthquake Recovery: Hazards - AIHA, accessed on September 12, 2025, https://www.aiha.org/public-resources/healthierworkplaces/healthier-community-resources/disaster-response-resource-center/health-and-safety-issues-in-natural-disasters/earthquake-recovery-hazards
40. National Disaster Recovery Framework (NDRF) Recovery Planning Through the Lens of Preparedness - Readiness and Emergency Management for Schools Technical Assistance Center, accessed on September 12, 2025, https://rems.ed.gov/docs/NDRFRecoveryPlanningFactSheet_508C.pdf
41. About NDRF – The National Disaster Recovery Framework - Denham Strong, accessed on September 12, 2025, https://denhamstrong.com/about-ndrf/
42. Types of Disasters | Teton County, WY, accessed on September 12, 2025, https://www.tetoncountywy.gov/404/Types-of-Disasters
43. What are the two types of floods? | U.S. Geological Survey - USGS.gov, accessed on September 12, 2025, https://www.usgs.gov/faqs/what-are-two-types-floods
44. Flood Types - Severe Weather 101, accessed on September 12, 2025, https://www.nssl.noaa.gov/education/svrwx101/floods/types/
45. Types & Causes of Flooding - About - OPW Flood Risk Management, accessed on September 12, 2025, https://www.floodinfo.ie/about_frm/types_and_causes/
46. Impacts of Flooding - About - OPW Flood Risk Management, accessed on September 12, 2025, https://www.floodinfo.ie/about_frm/impacts_of_flooding/
47. Flood impacts - Washington State Department of Ecology - | WA.gov, accessed on September 12, 2025, https://ecology.wa.gov/air-climate/responding-to-climate-change/flood-impacts
48. The Dangers of Floodwater | Johns Hopkins | Bloomberg School of Public Health, accessed on September 12, 2025, https://publichealth.jhu.edu/2024/the-dangers-of-floodwater
49. Flood Safety Tips - Mass.gov, accessed on September 12, 2025, https://www.mass.gov/info-details/flood-safety-tips
50. Recovery: What to do after a flood » National Emergency ..., accessed on September 12, 2025, https://www.civildefence.govt.nz/cdem-sector/consistent-messages/flood/recovery-what-to-do-after-a-flood
51. Flood forecasting systems | Climate Technology Centre & Network ..., accessed on September 12, 2025, https://www.ctc-n.org/technology-library/early-warning-systems-forecasting/flood-forecasting-systems
52. Flood Monitoring Systems Real Time Warning for Water Level and Flow - YSI, accessed on September 12, 2025, https://www.ysi.com/applications/flooding
53. Flood Forecasting - Google Research, accessed on September 12, 2025, https://sites.research.google/gr/floodforecasting/
54. Preparing for Floods - Centers for Disease Control and Prevention | CDC, accessed on September 12, 2025, https://www.cdc.gov/floods/safety/index.html
55. Flood Safety | Flood Preparedness | American Red Cross, accessed on September 12, 2025, https://www.redcross.org/get-help/how-to-prepare-for-emergencies/types-of-e

mergencies/flood.html
56. What To Do Before, During, and After a Flood | Patterson, CA - Official Website, accessed on September 12, 2025, https://www.pattersonca.gov/488/What-To-Do-Before-During-and-After-a-Flo
57. Help your community reduce flood risk and prepare for a flood emergency - First Street™, accessed on September 12, 2025, https://help.firststreet.org/hc/en-us/articles/360050401554-Help-your-community-reduce-flood-risk-and-prepare-for-a-flood-emergency
58. Be Informed: Community | Be Ready Utah, accessed on September 12, 2025, https://beready.utah.gov/community-preparedness/be-informed-community/
59. The Most Common Natural Disasters Across The U.S. - American Red Cross, accessed on September 12, 2025, https://www.redcross.org/get-help/how-to-prepare-for-emergencies/common-natural-disasters-across-us.html
60. Punjab: NDRF teams provide relief materials in flood-affected Fazilka, rescue 1500 villagers, accessed on September 12, 2025, https://www.aninews.in/news/national/general-news/punjab-ndrf-teams-provide-relief-materials-in-flood-affected-fazilka-rescue-1500-villagers20250905181232
61. Gujarat govt launches relief and rescue operations in rain-hit districts, accessed on September 12, 2025, https://timesofindia.indiatimes.com/city/ahmedabad/gujarat-govt-launches-relief-and-rescue-operations-in-rain-hit-districts/articleshow/123811891.cms
62. Army conducts disaster relief operations in flood-affected regions, accessed on September 12, 2025, https://timesofindia.indiatimes.com/city/chandigarh/army-conducts-disaster-relief-operations-in-flood-affected-regions/articleshow/123772469.cms
63. Flood Water Rescue (FWR) Equipment | NDRF - National Disaster ..., accessed on September 12, 2025, https://www.ndrf.gov.in/en/equipment/flood-water-rescue-fwr-equipment
64. The Ultimate Guide to Flood and Water Rescue Equipment in 2024, accessed on September 12, 2025, https://didiokmaking.com/blogs/news/the-ultimate-guide-to-flood-and-water-rescue-equipment-in-2024
65. RescueTECH | Water Rescue Equipment, accessed on September 12, 2025, https://www.rescuetech1.com/water-rescue-equipment.aspx
66. Allied Organizations | NDM India - Ministry of Home Affairs, accessed on September 12, 2025, https://ndmindia.mha.gov.in/ndmi/allied-organizations
67. National Disaster Management Authority (India) - PreventionWeb.net, accessed on September 12, 2025, https://www.preventionweb.net/organization/national-disaster-management-authority-india
68. National Disaster Management Authority (India) - Wikipedia, accessed on September 12, 2025, https://en.wikipedia.org/wiki/National_Disaster_Management_Authority_(India)
69. Kochi gets 222 crore boost to tackle city flooding crisis, accessed on September

12, 2025,
https://timesofindia.indiatimes.com/city/kochi/kochi-gets-222-crore-boost-to-tackle-city-flooding-crisis/articleshow/123683689.cms
70. Home | NDMA, GoI, accessed on September 12, 2025, https://www.ndma.gov.in/
71. Flood Relief - unicef usa, accessed on September 12, 2025, https://www.unicefusa.org/what-unicef-does/emergency-response/weather-disasters/floods
72. How to recover from a flood quickly and safely - The National Flood Insurance Program, accessed on September 12, 2025, https://www.floodsmart.gov/recover/checklist
73. Flood Recovery Booklet - ICPC, accessed on September 12, 2025, https://hosted.lib.uiowa.edu/flood/nine_steps.html
74. Cyber-attacks on critical health infrastructure - World Health Organization (WHO), accessed on September 12, 2025, https://www.who.int/news-room/questions-and-answers/item/cyber-attacks-on-critical-health-infrastructure
75. 10 Common Types of Cyberattacks and How to Prevent Them - Coursera, accessed on September 12, 2025, https://www.coursera.org/articles/types-of-cyber-attacks
76. What is a Cyber Attack? Types, Effects & Prevention - Fortinet, accessed on September 12, 2025, https://www.fortinet.com/resources/cyberglossary/what-is-cyber-attack
77. What is a Cyberattack? | IBM, accessed on September 12, 2025, https://www.ibm.com/think/topics/cyber-attack
78. What is Incident Response? - IBM, accessed on September 12, 2025, https://www.ibm.com/think/topics/incident-response
79. Top 20 Most Common Types Of Cyber Attacks | Fortinet, accessed on September 12, 2025, https://www.fortinet.com/resources/cyberglossary/types-of-cyber-attacks
80. What Is Cybersecurity? - IBM, accessed on September 12, 2025, https://www.ibm.com/think/topics/cybersecurity
81. Secure Cyberspace and Critical Infrastructure | Homeland Security, accessed on September 12, 2025, https://www.dhs.gov/archive/secure-cyberspace-and-critical-infrastructure
82. Cyber attacks on critical infrastructure - Allianz Commercial, accessed on September 12, 2025, https://commercial.allianz.com/news-and-insights/expert-risk-articles/cyber-attacks-on-critical-infrastructure.html
83. Top 6 Cyber Attack Prevention Strategies in 2025 - Cynet, accessed on September 12, 2025, https://www.cynet.com/advanced-threat-protection/top-6-cyber-attack-prevention-strategies-in-2025/
84. Cybersecurity | Ready.gov, accessed on September 12, 2025, https://www.ready.gov/cybersecurity
85. What Is Cyber Hygiene? Introduction, Best Practices, & Next Steps ..., accessed on

September 12, 2025,
https://www.splunk.com/en_us/blog/learn/cyber-hygiene.html
86. Protect your company from cyber attacks - Mass.gov, accessed on September 12, 2025,
https://www.mass.gov/info-details/protect-your-company-from-cyber-attacks
87. 12 Tips for Mitigating Cyber Risk | JPMorgan Chase, accessed on September 12, 2025,
https://www.jpmorgan.com/insights/cybersecurity/ransomware/12-tips-for-mitigating-cyber-risk
88. National Security Agency | Central Security Service, accessed on September 12, 2025, https://www.nsa.gov/
89. Cybersecurity - Homeland Security, accessed on September 12, 2025,
https://www.dhs.gov/topics/cybersecurity
90. carnegieendowment.org, accessed on September 12, 2025,
https://carnegieendowment.org/research/2025/03/interpreting-indias-cyber-statecraft?lang=en#:~:text=CERT%2DIn%20and%20the%20Cyber,and%20Information%20Technology%20(MeitY).
91. Government of India Taking Measures to Protect Critical ... - PIB, accessed on September 12, 2025,
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2116341
92. Government Strengthens Cybersecurity Across Critical Sectors; Over 9700 CERT-In Audits Conducted in 2024–25 - PIB, accessed on September 12, 2025,
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2148943
93. ENISA: Home, accessed on September 12, 2025, https://www.enisa.europa.eu/
94. Methods and Technologies to Detect Cyber Attacks - EC-Council University, accessed on September 12, 2025,
https://www.eccu.edu/blog/methods-technologies-detect-cyber-attacks/
95. Review of Cyberattack Implementation, Detection, and Mitigation Methods in Cyber-Physical Systems - MDPI, accessed on September 12, 2025,
https://www.mdpi.com/1996-1073/16/13/5206
96. What Is an Incident Response Plan (IRP)? - Palo Alto Networks, accessed on September 12, 2025,
https://www.paloaltonetworks.com/cyberpedia/incident-response-plan
97. Incident Response Plan: Frameworks and Steps | CrowdStrike, accessed on September 12, 2025,
https://www.crowdstrike.com/en-us/cybersecurity-101/incident-response/incident-response-steps/
98. The 6 Phases of an Incident Response Plan - Check Point Software, accessed on September 12, 2025,
https://www.checkpoint.com/cyber-hub/cyber-security/what-is-incident-response/the-6-phases-of-an-incident-response-plan/
99. www.undrr.org, accessed on September 12, 2025,
https://www.undrr.org/understanding-disaster-risk/terminology/hips/so0103#:~:text=A%20suggested%20definition%20for%20civil,protests%2C%20demonstrations%2C%20etc.)

100. Civil Unrest (SO0103) | UNDRR, accessed on September 12, 2025, https://www.undrr.org/understanding-disaster-risk/terminology/hips/so0103
101. 4.3.11 Civil Disorder, accessed on September 12, 2025, https://www.essexsheriff.com/wp-content/uploads/2019/12/Draft-Section-4.3.11-Civil-Disorder.pdf
102. Civil Disturbance - FL SHMP, accessed on September 12, 2025, https://flshmp-floridadisaster.hub.arcgis.com/pages/civil-disturbance
103. Explained: The Six Triggers that Spark Civil Unrest - Byline Times, accessed on September 12, 2025, https://bylinetimes.com/2024/08/29/six-factors-driving-civil-unrest/
104. Civil war - Wikipedia, accessed on September 12, 2025, https://en.wikipedia.org/wiki/Civil_war
105. Civil war | Definition, Causes, & Theories - Britannica, accessed on September 12, 2025, https://www.britannica.com/topic/civil-war
106. Causes Of The Civil War | History Detectives - PBS, accessed on September 12, 2025, https://www.pbs.org/opb/historydetectives/feature/causes-of-the-civil-war/
107. Trigger Events of the Civil War | American Battlefield Trust, accessed on September 12, 2025, https://www.battlefields.org/learn/articles/trigger-events-civil-war
108. BLOOD AND METTLE: THE EFFECTS OF CIVIL UNREST ON ..., accessed on September 12, 2025, https://digitalcommons.spu.edu/cgi/viewcontent.cgi?article=1243&context=honorsprojects
109. The cost of violence: Estimating the economic impact of conflict - International Growth Centre, accessed on September 12, 2025, https://www.theigc.org/sites/default/files/2016/12/IGCJ5023_Economic_Cost_of_Conflict_Brief_2211_v7_WEB.pdf
110. Publication: Unemployment and Participation in Violence - Open Knowledge Repository, accessed on September 12, 2025, https://openknowledge.worldbank.org/entities/publication/10350600-3736-55e5-8ce3-5e41e20c90ff
111. The Role of Social Media in Protest Participation: The Case of Candlelight Vigils in South Korea - International Journal of Communication, accessed on September 12, 2025, https://ijoc.org/index.php/ijoc/article/viewFile/7767/2313
112. Social media: A tool for peace or conflict? | SIPRI, accessed on September 12, 2025, https://www.sipri.org/commentary/topical-backgrounder/2021/social-media-tool-peace-or-conflict
113. The Role of Social Media in Mobilizing Political Protest - German Institute of Development and Sustainability (IDOS), accessed on September 12, 2025, https://www.idos-research.de/uploads/media/DP_10.2012.pdf
114. Social Media Effects: Hijacking Democracy and Civility in Civic Engagement - PMC, accessed on September 12, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC7343248/

115. Future Protest Made Risky: Examining Social Media Based Civil Unrest Prediction Research and Products - PubMed Central, accessed on September 12, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC8423833/
116. Civil Unrest Resources - EMS.gov, accessed on September 12, 2025, https://www.ems.gov/assets/Guidance_Resources_Civil_Unrest.pdf
117. Civil Unrest: Travel Advice and Planning | Harvard GSS, accessed on September 12, 2025, https://www.globalsupport.harvard.edu/travel/advice/civil-unrest-planning
118. Staying Safe During Protests, Civil Unrest and Riots - Solace Global, accessed on September 12, 2025, https://www.solaceglobal.com/report/civil-unrest-guide/
119. Toolkit: Protest safety tips from Greenpeace, accessed on September 12, 2025, https://www.greenpeace.org/usa/protest-safety-tips-from-greenpeace/
120. Security Advice: Civil Unrest, accessed on September 12, 2025, https://www.ucop.edu/risk-services-travel/_files/security-advice-civil-unrest.pdf
121. Find ways to de-escalate intense interactions - GovDelivery, accessed on September 12, 2025, https://content.govdelivery.com/accounts/MNADMIN/bulletins/3d754b0
122. 7 De-Escalation Skills Essential for Defusing Conflict | Pollack ..., accessed on September 12, 2025, https://pollackpeacebuilding.com/blog/de-escalation-skills/
123. The 3 Stages of De-Escalation & How to Use Each to Your Favor - Defuse, accessed on September 12, 2025, https://deescalation-training.com/2024/10/stages-of-de-escalation/
124. Who Polices During Riots and Protests? - Nolo, accessed on September 12, 2025, https://www.nolo.com/legal-encyclopedia/policing-and-civil-unrest.html
125. 12 things every police department's civil unrest plan needs - Police1, accessed on September 12, 2025, https://www.police1.com/police-products/emergency-preparedness/articles/12-things-every-police-departments-civil-unrest-plan-needs-Nny9wOSfXLUIYBps/
126. Managing Violent Civil Disturbances - FDLE, accessed on September 12, 2025, https://www.fdle.state.fl.us/FCJEI/Programs/SLP/Documents/Full-Text/Simovich.aspx
127. In post-Jan. 6 era, 600 officers train for riots as threats to lawmakers climb toward record high, accessed on September 12, 2025, https://apnews.com/article/capitol-police-jan-6-secret-service-riot-washington-ef20e1075fd210e697aa0a15f0fd57aa
128. Post-Conflict Economic Reconstruction | The Princeton Encyclopedia of Self-Determination, accessed on September 12, 2025, https://pesd.princeton.edu/node/586
129. Rebuilding Societies After Conflict: Teaching Guide - United States Institute of Peace, accessed on September 12, 2025, https://www.usip.org/sites/default/files/rebuilding.pdf
130. Community-led Approach: The Foundation of Peacebuilding - UNITAR, accessed on September 12, 2025, https://unitar.org/community-led-approach-foundation-peacebuilding
131. Community-led peacebuilding | PCi - Peaceful Change Initiative, accessed on

September 12, 2025, https://peacefulchange.org/our-work/community-led-peacebuilding/
132. Peace Building in Post-Conflict Societies: Processes and Strategies, accessed on September 12, 2025, https://www3.gmu.edu/programs/icar/hsp/hj.html
133. Community-based approaches to peacebuilding in conflict-affected and fragile contexts, accessed on September 12, 2025, https://inee.org/sites/default/files/Community_Based_Approaches_Peacebuilding.pdf
134. Reconciliation After Violent Conflict - International IDEA, accessed on September 12, 2025, https://www.idea.int/sites/default/files/publications/chapters/reconciliation-after-violent-conflict/reconciliation-after-violent-conflict-policy-summary.pdf
135. 2020 Delhi riots - Wikipedia, accessed on September 12, 2025, https://en.wikipedia.org/wiki/2020_Delhi_riots
136. Violence against Muslims in India - Wikipedia, accessed on September 12, 2025, https://en.wikipedia.org/wiki/Violence_against_Muslims_in_India
137. Ethnic cleansing and brewing civil war in the case of India's Manipur - LSE Human Rights, accessed on September 12, 2025, https://blogs.lse.ac.uk/humanrights/2024/03/14/ethnic-cleansing-and-brewing-civil-war-in-the-case-of-indias-manipur/
138. Sambhal: A Case Study in the Weaponization of History and Communal Violence, accessed on September 12, 2025, https://www.hindusforhumanrights.org/en/blog/sambhal-a-case-study-in-the-weaponization-of-history-and-communal-violence
139. Communal Riots and Its Psychological Impact: A Systematic Review Study in Indian Context, accessed on September 12, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11100963/
140. Party Stabilization, Declining Riot Violence, and New Modalities of Political Conflict in India (Chapter 6) - Playing with Fire - Cambridge University Press & Assessment, accessed on September 12, 2025, https://www.cambridge.org/core/books/playing-with-fire/party-stabilization-declining-riot-violence-and-new-modalities-of-political-conflict-in-india/75C3BF638C8709A4E1E7B324D8D3C487